

BEYA, Russian Federation

WMO: 299620

Lat: 53.050N

Lon: 90.917E

Elev: 1536

StdP: 13.90

Time Zone: 7.00 (E07)

Period: 86-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.0	(c) -17.0	(d) -27.3	(e) 1.3	(f) -23.1	(g) -21.7	(h) 1.8	(i) -16.8	(j) 26.6	(k) 13.9	(l) 20.9	(m) 12.0	(n) 0.9	(o) 180	(p) 0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	83.2	64.0	80.1	63.5	77.1	62.1	67.1	78.9	65.4	76.7	63.7	74.3	5.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	90.2	73.0	61.0	84.7	71.1	59.3	79.9	69.6	32.6	78.9	31.2	77.1	29.9	74.3	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.4	(b) 16.4	(c) 14.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	5.2	11.4	23.3	38.9	52.7	60.9	64.8	61.5	47.9	36.7	18.5	13.5
	DBStd	22.72	12.37	11.17	12.32	8.45	8.80	7.19	5.48	5.88	7.32	8.86	13.46	11.28
	HDD50	6345	1390	1081	828	342	76	7	0	1	124	419	945	1132
	HDD65	10575	1855	1501	1292	782	391	160	69	140	514	879	1395	1597
	CDD50	1388	0	0	0	10	159	335	460	357	61	6	0	0
	CDD65	140	0	0	0	0	8	38	63	30	1	0	0	0
	CDH74	1343	0	0	0	0	153	350	516	318	6	0	0	0
	CDH80	272	0	0	0	0	30	65	107	70	0	0	0	0
(14) Wind	WSAvg	5.0	3.8	4.7	4.9	6.5	5.9	5.0	3.9	4.0	4.8	5.4	5.6	5.2
(15) Precipitation	PrecAvg	14.4	0.3	0.2	0.4	0.8	1.5	2.7	2.9	2.6	1.4	0.7	0.6	0.4
	PrecMax	22.0	0.7	0.6	1.3	1.9	4.3	5.4	4.4	4.5	3.1	2.3	1.4	0.7
	PrecMin	10.1	0.0	0.0	0.1	0.1	0.4	1.1	1.7	1.1	0.5	0.2	0.2	0.1
	PrecStd	2.5	0.2	0.1	0.3	0.5	0.8	1.1	0.7	1.1	0.6	0.4	0.4	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	36.9	58.6	69.2	83.8	85.9	87.7	86.6	73.6	63.0	47.6	40.0
		MCWB	28.6	30.9	45.4	49.7	59.9	63.5	67.3	65.9	57.5	48.6	37.9	35.1
	2%	DB	29.7	33.3	48.8	64.2	78.6	81.6	82.7	81.5	68.6	58.8	41.8	35.0
		MCWB	25.8	28.4	39.5	47.6	57.2	62.6	66.3	64.3	55.3	46.6	35.2	30.5
	5%	DB	25.6	30.3	42.8	60.0	74.3	78.1	79.7	77.4	65.3	55.0	39.1	31.8
		MCWB	22.3	26.1	35.3	45.9	55.6	62.0	65.0	63.0	54.0	45.4	33.8	27.8
	10%	DB	22.6	27.7	38.2	54.9	70.0	74.9	76.6	73.8	61.0	50.0	36.2	28.9
		MCWB	19.5	23.9	32.0	43.6	53.8	60.8	63.7	61.7	51.7	42.5	31.6	25.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.1	31.3	45.7	52.1	62.4	67.8	69.6	68.6	60.8	52.7	39.1	35.8
		MCDB	33.0	35.9	57.5	66.6	79.7	79.1	81.0	80.9	69.9	60.3	45.8	39.7
	2%	WB	25.9	28.7	40.2	48.7	59.0	65.1	67.5	66.2	57.4	48.5	36.2	30.8
		MCDB	29.5	32.8	48.6	61.4	74.1	77.3	79.7	77.8	66.1	56.3	41.1	34.6
	5%	WB	22.6	26.4	36.1	46.5	57.0	63.1	66.2	64.2	54.8	45.6	34.0	28.0
		MCDB	25.5	30.4	42.4	57.9	71.2	75.4	77.8	75.6	62.9	53.6	39.1	31.7
	10%	WB	19.7	23.8	32.4	44.0	54.8	61.5	64.6	62.3	52.3	42.8	31.6	25.1
		MCDB	22.5	27.6	37.7	54.7	67.9	73.2	74.7	72.0	59.8	50.0	36.3	28.8
(35) Mean Daily Temperature Range	MDBR		15.4	16.4	17.3	19.2	20.8	18.0	17.2	18.6	16.7	14.9	13.9	13.8
	5% DB	MCDBR	18.8	18.6	21.0	28.1	28.8	24.0	23.6	25.8	23.9	22.3	16.2	16.5
		MCWBR	15.9	15.5	13.4	15.6	12.8	10.5	10.1	12.3	12.5	12.6	11.6	14.2
	5% WB	MCDBR	18.2	18.2	20.1	24.8	23.7	21.1	21.4	22.2	20.6	21.0	15.2	16.8
		MCWBR	16.1	15.3	13.4	14.5	11.1	10.5	9.9	11.4	12.1	13.1	11.1	14.9
(40) Clear-Sky Solar Irradiance	taub		0.280	0.316	0.365	0.409	0.413	0.400	0.408	0.394	0.357	0.345	0.319	0.285
	taud		2.074	2.023	2.016	2.135	2.180	2.319	2.320	2.331	2.372	2.272	2.125	2.030
	Ebn at Noon		212	240	253	259	265	270	265	262	259	232	193	180
	Edh at Noon		23	34	43	43	43	38	38	35	30	27	22	20
(44) All-Sky Solar Radiation	RadAvg		346	675	1127	1436	1682	1915	1746	1472	1063	605	314	227
	RadStd		26	46	82	88	146	161	117	115	94	56	36	23

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page